Fiscal Year:	FY 2024	**Task Last Updated:**	FY 05/29/2024
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Mechanical and Gravitational Countermeasures to Ocular Changes During Strict Head-Down Tilt Bedrest		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Johnson Space Center Cardiovascular and Vision Laboratory		
PI Address 2:	2101 NASA Parkway, HAC/B21N-1207		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/26/2022	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per C. Ribeiro/NASA-JSC. (Ed., 11/27/23). Original end date was 08/20/2024. NOTE: Project start date updated due to P.I. change to B. Macias. Information per C. Ribeiro/NASA-JSC. (Ed. 8/29/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Huang, Alex M.D., Ph.D. (Doheny Eye Institute) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Lovering, Andrew Ph.D. (University of Oregon) Martin, Bryn Ph.D. (Alcyone Therapeutics Inc.) Young, Millennia Ph.D. (NASA Johnson Space Center) Lytle, Jason (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The Spaceflight Associated Neuro-ocular Syndrome (SANS) is associated with structural and functional ocular and brain changes and develops in ~2/3rds of astronauts during long-duration spaceflight. Although the etiology of SANS is unknown, a weightlessness-driven headward fluid redistribution relative to the upright position on Earth is hypothesized to be a primary initiating factor. A countermeasure that can successfully reverse this headward fluid shift for multiple hours per day may be needed to prevent SANS. Exposure to 30 days of the spaceflight analog strict 6 degree head-down tilt (HDT) bedrest induces the development of optic disc edema, a hallmark sign of SANS, in 2/3 of subjects; thus, NASA will utilize this platform to test the effectiveness of various countermeasures to prevent or reduce the development of SANS findings. This study will include a control group exposed to 30 days of strict 6 degree HDT. A second group of subjects will be exposed to 6 hours of 25 mmHg lower body negative pressure (LBNP), daily, during 30 days of strict 6 degree HDT bedrest. A third group of subjects will be exposed to 6 hours of upright posture per day, and 6 degree HDT for the remaining 18 h/day. This will help determine if daily exposure to 6 hours total of upright posture, fully reversing the headward fluid for the same duration of use as the LBNP group, is sufficient to prevent SANS findings, and thus can direct future countermeasure studies with respect to the duration needed to successfully prevent SANS. We will assess multiple cerebral, vascular, and ocular outcome measures to determine the effects of 30 days strict 6 degree HDT bedrest, and the ability of daily 6 hour exposure to LBNP or upright positioning to prevent these changes. Specific Aims: 1. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents ocular and cerebral structural changes during 30 days of strict 6° head-down tilt bedrest. 2. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents ocular functional changes during 30 days of strict 6° head-down tilt bedrest. 3. To determine if daily exposure to actual (upright posture) and simulated (LBNP) hydrostatic pressure gradients prevents vascular and cerebrospinal fluid changes during 30 days of strict 6° head-down tilt bedrest. This bedrest study will be implemented in the :envihab facility at the German Aerospace Center (DLR) in Cologne, Germany.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spaceflight associated neuro-ocular syndrome (SANS) develops in approximately 2/3 of astronauts during long-duration spaceflight. Although the etiology of SANS is unknown, a weightlessness-driven headward fluid redistribution relative to the upright position on Earth is hypothesized to be a primary initiating factor. A countermeasure that can successfully reverse this headward fluid shift for multiple hours per day may be needed to prevent SANS findings from developing. Exposure to 30 days of strict 6-degree head-down tilt (HDT) bedrest induces the development of optic disc edema, a hallmark sign of SANS, in approximately 2/3 of subjects; thus, NASA is utilizing this platform to test the effectiveness of various countermeasures to prevent or reduce the development of SANS findings.
Task Progress:	Summary of Key Updates This Period • Data collection was completed for campaigns 3 and 4. • All data were transferred to the NASA Johnson Space Center (JSC) Cardiovascular and Vision Laboratory and analyses have been completed. • A manuscript reporting on the effectiveness of the lower body negative pressure (LBNP) countermeasure (C1 and C2) will be submitted to J Appl Physiol in Spring 2024. • A manuscript reporting on the effectiveness of Exercise + Thigh Cuff countermeasure is being drafted also for submission to J Appl Physiol. • A manuscript summarizing eye findings across all campaigns is being led by Dr. Alex Huang and is in the early stages of development. • All equipment that was shipped to German Aerospace Center (DLR) to support data collection has been returned to NASA, except for certain equipment planned for use in upcoming NASA bedrest studies at DLR.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Marshall-Goebel K, Laurie SS, Pardon LP, Greenwald SH, Jasion J, Lytle J, Martin D, Lee SMC, Lovering AT, Huang AS, Martin BA, Brunstetter T, Young M, Kramer LA, Macias BR. "Mechanical and gravitational countermeasures to ocular changes during strict head-down tilt bedrest." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop (2022), Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Marshall-Goebel K, Lytle JR, Laurie SS, Lee SMC, Pardon LP, Martin D, Young M, Hargens AR, Sargsyan A, Ebert D, Dulchavsky S, Macias BR. "Countermeasures targeting reversal of the weightlessness-induced headward fluid shift to mitigate spaceflight associated neuro-ocular syndrome." 2022 Aerospace Medical Association Annual Scientific Meeting, Reno, Nevada, May 22-26, 2022. Abstracts. 2022 Aerospace Medical Association Annual Scientific Meeting, Reno, Nevada, May 22-26, 2022. , May-2022
Articles in Peer-reviewed Journals	Moestl S, De Boni L, Hoenemann J-N, Kramer T, Schmitz J, Pesta D, Frett T, Bohmeier M, Frings-Meuthen P, Ewald AC, Nitsche A, Loehr P, Noppe A, Klischies N, Huang AS, Laurie SS, Marshall-Goebel K, Macias BR, Tank J, Jordan J, Mulder E. "Mechanical countermeasures for spaceflight-associated neuro-ocular syndrome during 30-days of head-down tilt bedrest: Design, implementation, and tolerability." Front Physiol. 2025 Feb 23;16:1530783. https://doi.org/10.3389/fphys.2025.1530783 ; PMID: 40066284; PMCID: PMC11891356 , Feb-2025